

Methods for Multiple Change Point Detection

Tara Welch

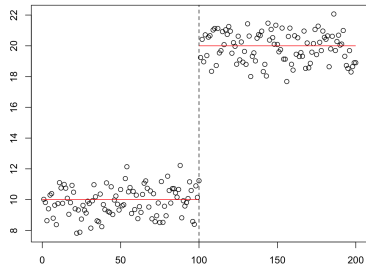
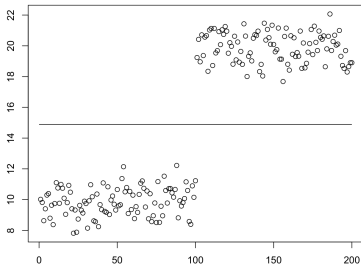
Durham University

25 February 2026

Motivation

- Real-world data is inherently noisy.
- How can we distinguish between stochastic noise and a change in the underlying process?
- Statistical analysis largely assumes **stationarity**.

Motivation



- Failing to identify the change can impact the analysis.

Time Series Data

Time Series Sequence of data points at successive discrete time intervals, $\{X_t : t \in 1, \dots, n\}$.

Change Point A discontinuity in a time series where the distribution changes.

Model Assumptions

$$X_t = f(t) + \varepsilon_t$$

Underlying Signal

Noise

$\varepsilon_t \stackrel{i.i.d}{\sim} \mathcal{N}(0, \sigma^2)$

Types of Change Points

- Time series data modelled as **piecewise stationary**.
- Parameter change:
 - Change in mean
 - Change in variance
- Change in distribution

Types of Change Points

- Time series data modelled as **piecewise stationary**.
- Parameter change:
 - **Change in mean**

$$X_t \sim \begin{cases} \mathcal{N}(\mu_1, \sigma), & \text{for } 1 \leq t \leq \tau \\ \mathcal{N}(\mu_2, \sigma), & \text{for } \tau + 1 \leq t \leq n \end{cases}$$

Detecting a Single Change Point

$\mathcal{H}_0$: There is no change point.

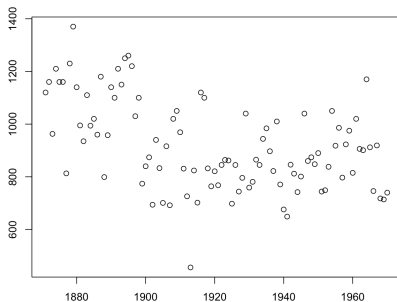
$\mathcal{H}_1$: There exists a change point at $t = \tau$.

Generalised Log-Likelihood Ratio (GLLR) Test:

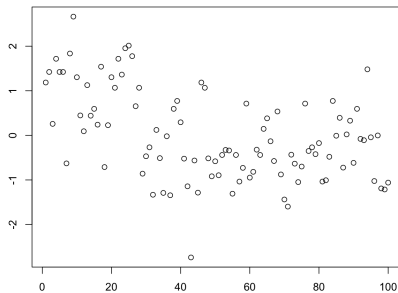
$$\mathcal{LR}_t = 2 \left[\mathcal{L}(\hat{\theta}_1; X_{1:t}) + \mathcal{L}(\hat{\theta}_2; X_{(t+1):n}) - \mathcal{L}(\hat{\theta}_0; X_{1:n}) \right]$$

$$\hat{\tau} = \arg \max_{t=1, \dots, n-1} \mathcal{LR}_t$$

Single Change in Mean Example



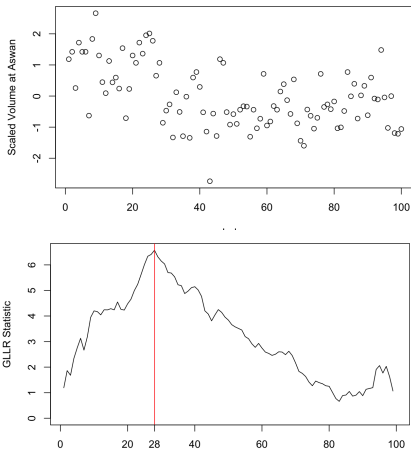
(a) Volume of Nile river at Aswan from 1871-1970. From R package "datasets".



(b) Shifted and scaled Nile data

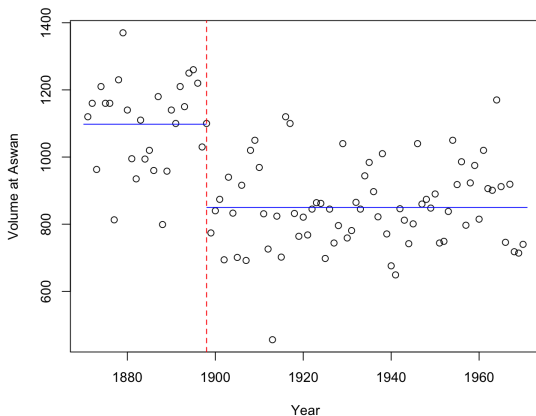
- Scale the data by the global mean and variance.
- Shift the x-axis so that the first data point has index 1.

Single Change in Mean Example



- Calculate GLLR statistic for each $t \in \{1, n - 1\}$.
- Maximum at $t = 28$.

Single Change in Mean Example



- Visualise on original data. $t = 28$ corresponds to 1898.
- A dam was built in 1898, indicates “true” change.

Choosing a Threshold

- We declare a change if $\mathcal{LR}_\tau > c$, for some threshold c .

Hannan-Quinn Information Criterion (HQIC) $4 \log \log n$	Bayes Information Criterion (BIC) $2 \log n$
Derived from tail behaviour of normal distribution	Derived from posterior probabilities
Good for subtle changes	Good for filtering noise

Choosing a Threshold

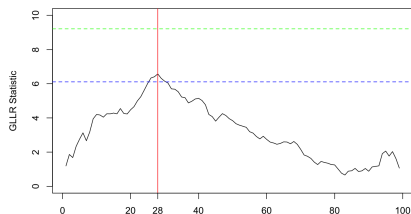


Figure: GLLR statistic for Nile data with both thresholds shown.

HQIC ≈ 6.11 , BIC ≈ 9.21

- Small change, noisy data $\rightarrow$ BIC does not detect change.
- Normal assumptions hold $\rightarrow$ use HQIC.

Multiple Change Point Example

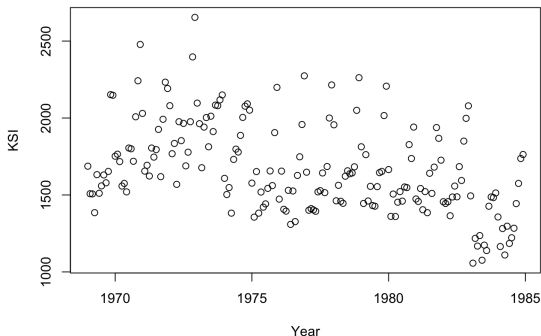


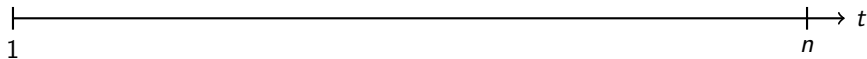
Figure: Number of drivers killed or seriously injured (KSI) in the UK between Jan 1969 and Dec 1984. From the R package “datasets”.

Methods for Detecting Multiple Change Points

- Top-down
- Bottom-up
- Sliding window

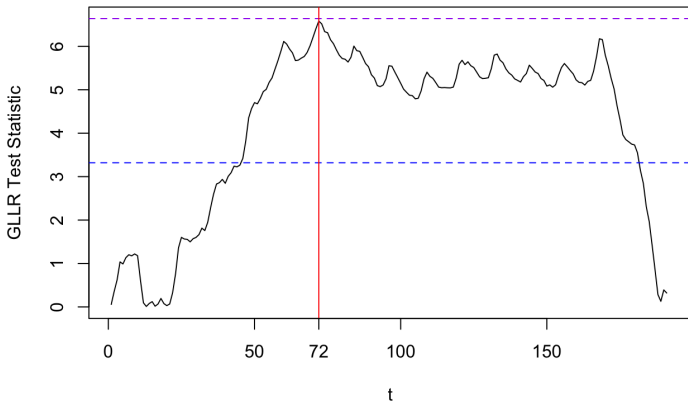
Binary Segmentation

Full Dataset
 $t \in \{1, \dots, n\}$



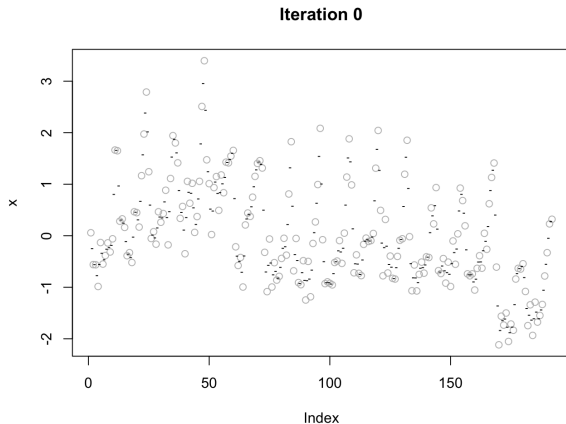
Binary Segmentation

1 to 192



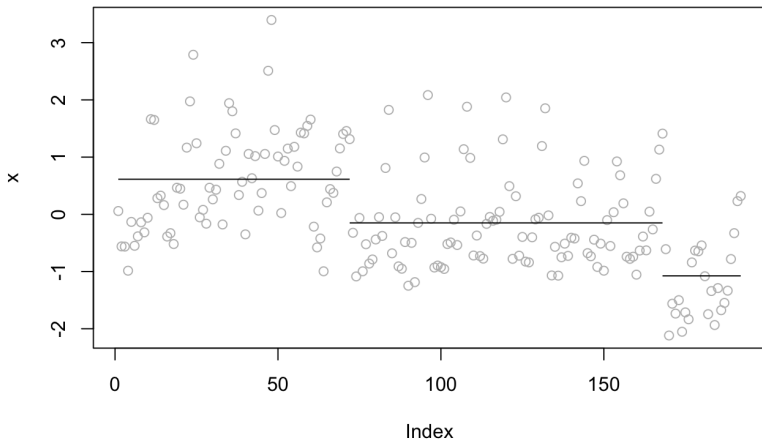
Thresholds: $4 \log \log n$ does not detect any changes.
 $2 \log \log n$ used (smaller, strongly consistent).

Bottom-Up



Bottom-Up Results

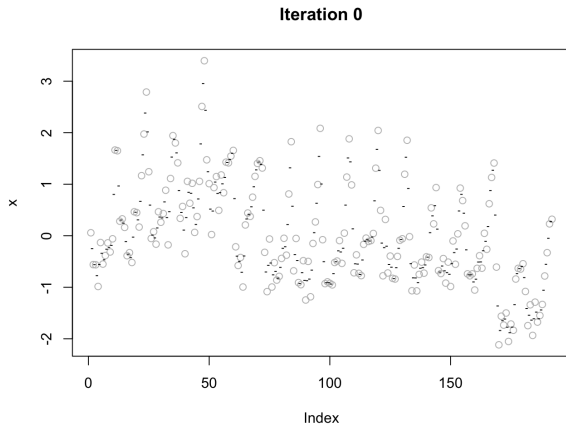
Iteration 189



Bottom-Up Threshold

- Seatbelt data is noisy.
- With HQIC, 6 changes were detected → too sensitive.
- BIC used.

Sliding Window



Sliding Window Summary

Complexity:	$O(n)$
Threshold Choice:	$4 \log \log n$ (HQIC)
Advantage:	Fast, local, can be used online
Disadvantage:	Highly dependent on choice of h

Comparison of Methods

Method	Change Point Locations
Binary Segmentation	10-1969, 02-1975, 01-1983
Bottom-Up	01-1975, 01-1983
Sliding Window	09-1973, 01-1975, 02-1983

Two major real-world changes occurred over this period:

- Seatbelts compulsory equipment in all new cars from 1972.
- Legal requirement for seatbelts to be worn from January 31st 1983.

Comparison of Methods

Method	Change Point Locations
Binary Segmentation	10-1969, 02-1975, 01-1983
Bottom-Up	01-1975, 01-1983
Sliding Window	09-1973, 01-1975, 02-1983

These change points are likely to be as a result of the addition of seatbelts to new cars.

Comparison of Methods

Method	Change Point Locations
Binary Segmentation	10-1969, 02-1975, 01-1983
Bottom-Up	01-1975, 01-1983
Sliding Window	09-1973, 01-1975, 02-1983

These change points are likely to be as a result of the legal requirement for seatbelts to be worn.

Summary

- All methods identified the two key changes.

Summary

- All methods identified the two key changes.
- These changes can be related to real-world events.
- Importance of threshold selection.

References I



R Core Team.

R: A Language and Environment for Statistical Computing.

R Foundation for Statistical Computing, 2024.



D. V. Hinkley.

Inference about the change-point in a sequence of random variables.

Biometrika, 57(1):1–17, 1970.



E. J. Hannan & B. G. Quinn.

The determination of the order of an autoregression.

Journal of the Royal Statistical Society. Series B (Methodological),
41(2):190–195, 1979.



G Schwarz.

Estimating the dimension of a model.

The Annals of Statistics, 6(2):461–464, 1978.

References II



A. J. Scott & M. Knott.

A cluster analysis method for grouping means in the analysis of variance.

Biometrics, 30(3):507–512, 1974.



P. Fryzlewicz.

Wild binary segmentation for multiple change-point detection.

The Annals of Statistics, 42(6):2243–2281, 2014.



E. J. Keogh & S. Chu & D. Hart & M. Pazzani.

An online algorithm for segmenting time series.

In *Proceedings - IEEE International Conference on Data Mining, ICDM*, pages 289–296, 02 2001.

References III



T. Levajković & M. Messer.

Multiscale change point detection via gradual bandwidth adjustment in moving sum processes.

Electronic Journal of Statistics, 17(1):70 – 101, 2023.